Adding and Subtracting Rational Expressions

Algebra 2 Lesson 4-4 (fast track) Name _____ Date _____ Period ____

Learning goal: I can find the sum or difference of rational expressions and write a compound fraction in a simpler form.
Language goal: I can explain each step of adding, subtracting or simplifying using the words factor, least common denominator, numerator and compound fraction.

On the quiz: Question 1 adds with different denominators: **Example 3**. Questions 2 (an LCM) and 3 (a subtraction) use its steps (full Savvas quiz only). Questions 4 and 5 simplify a compound fraction: **Example 6**. The short form grades 1 and 4 (10 points each); 5 is a +5 bonus.

On the Topic 4 test: Question 10 asks you to add $1/(x+3)$ and $-6/(x^2-9)$ (Example 3). Question 15 asks for $1/a$ when $a = 1/x + 1/y$ (Example 6, like the Practice block). Question 4 (subtract over $x-3$ and $3-x$) comes from Example 4, which this packet skips.

Example 3: Add Rational Expressions With Unlike Denominators

What is the sum of $\dfrac{x+3}{x^2-1}$ and $\dfrac{2}{x^2-3x+2}$? (Use the steps for adding number fractions.)

$$= \frac{x+3}{(x+1)(x-1)} + \frac{2}{(x-1)(x-2)}$$

1. Factor each denominator.

$$= \frac{(x+3)(x-2)}{(x+1)(x-1)(x-2)} + \frac{2(x+1)}{(x+1)(x-1)(x-2)}$$

2. Build the LCD.

$$= \frac{(x^2+x-6)+(2x+2)}{(x+1)(x-1)(x-2)} = \frac{x^2+3x-4}{(x+1)(x-1)(x-2)}$$

3. Add the numerators.

$$= \frac{(x+4)(x-1)}{(x+1)(x-1)(x-2)} = \frac{x+4}{(x+1)(x-2)}$$

4. Factor again; divide out $x-1$.

LCD: each different factor once, to its highest power ($x-1$ is in both, so it is used once). Excluded values come from the *original* denominators: **answer** $(x+4)/\big((x+1)(x-2)\big)$, $x \neq -1, 1, 2$.

Say it: First I _____ each denominator. The least common denominator is _____. I add the _____ and keep the LCD. Then I factor again and divide out _____.

Try It 3: Find the sum

a. $\dfrac{x+6}{x^2-4} + \dfrac{2}{x^2-5x+6}$

LCD = _____

_____

_____

_____

Answer: _____ for $x \neq$ _____

b. $\dfrac{2x}{3x+4} + \dfrac{4x^2-11x-12}{6x^2+5x-4}$

LCD = _____

_____

_____

_____

Answer: _____ for $x \neq$ _____

Example 6: Simplify a Compound Fraction

A **compound fraction** has a fraction in its numerator, its denominator, or both (*fracción compuesta*). Write a simpler form of the compound fraction

$$\frac{\frac{1}{x}+\frac{2}{x+1}}{\frac{1}{y}}.$$

1. List every small denominator: x, $x+1$, y. Their LCM (the least common denominator) is $x(x+1)y$.

2. Multiply the whole numerator and the whole denominator by $\frac{x(x+1)y}{1}$. That is multiplying by 1, so the value does not change.

$$\frac{\left[\frac{1}{x}+\frac{2}{x+1}\right]\cdot x(x+1)y}{\frac{1}{y}\cdot x(x+1)y}=\frac{\frac{1}{x}\cdot x(x+1)y+\frac{2}{x+1}\cdot x(x+1)y}{\frac{1}{y}\cdot x(x+1)y}$$

3. Divide out each small denominator. Every small fraction is gone:

$$=\frac{(x+1)y+2xy}{x(x+1)}$$

4. Simplify the numerator: $(x+1)y+2xy=xy+y+2xy=3xy+y=(3x+1)y$.

$$=\frac{(3x+1)y}{x(x+1)}$$

Answer: $\frac{(3x+1)y}{x(x+1)}$ when $x\neq -1$, 0 and $y\neq 0$ (no small denominator can be 0).

Say it: This is a _____ because its numerator has fractions in it. I multiply the top and the bottom by the _____, which is _____. Then no small fractions are left, and I simplify the _____.

Try It 6: Simplify each compound fraction

a. $\dfrac{\frac{1}{x-1}}{\frac{x+1}{3}+\frac{4}{x-1}}$

Small denominators: _____

LCD = _____

_____

_____

_____

_____

_____

Answer: _____ for $x\neq$ _____

b. $\dfrac{2-\frac{1}{x}}{x+\frac{2}{x}}$

Small denominators: _____

LCD = _____

_____

_____

_____

_____

_____

Answer: _____ for $x\neq$ _____

Practice: Plug into a formula, then simplify

A car drives to a town at x miles per hour and drives back at $x + 10$ miles per hour. Its average speed for the round trip is

$$s = \frac{2}{\frac{1}{x} + \frac{1}{x+10}}.$$

a. The small denominators are _____. The LCD is _____. Simplify s.

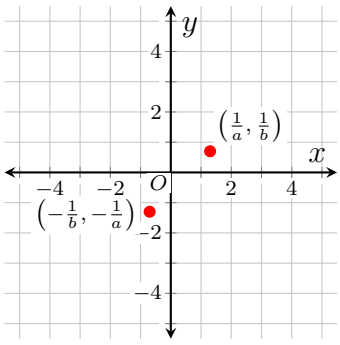
b. Use your answer: if $x = 40$, what is the average speed? Is it 45 mph?

IXL: Practice for the quiz

Code	Skill	Feeds
FEX ★	Add and subtract rational expressions	Quiz 1; Quiz 3 (full Savvas quiz only) (Example 3)
YD8 ★	Simplify complex fractions	Quiz 4 and 5 (Example 6)
UB5	Factor quadratics	Warm-up: step 1; Quiz 2 (LCM; full Savvas quiz only)
37N	Simplify rational expressions	Warm-up: the last step

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Use structure (Practice & Problem Solving 14)



Find the slope of the line that passes through the points shown. Express it in simplest form.

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = _____$$

Before the quiz, finish these sentences.

To add or subtract, I first _____ each denominator and find the _____.

After I add, I factor the new _____ and divide out a factor it shares with the denominator.

To simplify a _____, I multiply its top and bottom by the LCD of all the small fractions.